APPENDIX B

NOTABLE RANDOM VARIABLES AND THEIR PROPERTIES

This chapter recollects the properties of the random variables and random processes commonly used in the context on network design, some fundamental bounds on the probability distribution of random variables, and a few other classic results of probability theory.

—Andrea Zanella

B.1 Notable discrete random variables

B.2 Notable continuous random variables

B.3 Stochastic bounds

A first bound on the tail of the probability distribution of any non-negative r.v. is provided by the following Markov inequality.

Theorem B.1 (Markov inequality) *Let X be a non-negative random variable, with mean $\mathrm{m}_X = \mathrm{E}[X]$. Then, for any $a > 0$, it holds*

$$\mathrm{P}[X \geq a] \leq \frac{\mathrm{E}[X]}{a} = \frac{\mathrm{m}_X}{a} .$$

By Andrea Zanella Copyright © 2019 John Wiley & Sons, Inc.

Proof: Consider the r.v. Y defined as follows

$$Y = \begin{cases} 0\,, & \text{for } X < a\,; \\ a\,, & \text{for } X \geq a\,. \end{cases}$$

More compactly, we can write

$$Y = a \cdot \chi\left\{X \geq a\right\}\;;$$

where $\chi\left\{A\right\} = 1$ if A holds true, and $\chi\left\{A\right\} = 0$ otherwise. It is then clear that $Y \leq X$, i,e., $0 \geq Y - X$. Therefore, we also get $0 \geq \mathrm{E}\left[Y - X\right] = \int_0^\infty \mathrm{E}\left[Y - X | X = u\right] f_X(a)\mathrm{d}u$, where $f_X(u)$ is the PDF of X. Replacing Y with its value for a given $X = u$ we get $0 \geq \int_0^\infty \mathrm{E}\left[Y|X = u\right] f_X(u)\mathrm{d}u - \mathrm{E}\left[X\right] = \int_a^\infty a f_X(u)\mathrm{d}u - \mathrm{E}\left[X\right] = a\mathrm{P}\left[X \geq a\right] - \mathrm{E}\left[X\right]$. Since $a > 0$, we can hence write $\mathrm{P}\left[X \geq a\right] \leq \mathrm{E}\left[X\right]/a$, which concludes the proof. ∎

A generalization of the Markov inequality is obtained by considering monotonically increasing functions φ, which are positive on the values taken by the r.v..

Theorem B.2 (Markov monotonic inequality) *For any r.v. X and any monotonically increasing function φ which is non-negative for the non-negative reals, then, for any $a > 0$, it holds*

$$\mathrm{P}\left[|X| \geq a\right] \leq \frac{\mathrm{E}\left[\varphi(|X|)\right]}{\varphi(a)}\;.$$

Proof: We have

$$\mathrm{P}\left[|X| \geq a\right] = \mathrm{P}\left[\varphi(|X|) \geq \varphi(a)\right] \leq \frac{\mathrm{E}\left[\varphi(|X|)\right]}{\varphi(a)}$$

where the first inequality follows from the fact that φ is strictly monotonic, and the last derivation is given by the Markov inequality, which holds because φ is non-negative for non-negative real values. ∎

In particular, choosing $\varphi(a) = a^n$ for any $n > 0$, the Markov monotonic inequality yields

$$\mathrm{P}\left[|X| \geq a\right] \leq \frac{\mathrm{E}\left[|X|^n\right]}{a^n}\,. \tag{B.1}$$

Note that, if X is non-negative, or if n is even, then $\mathrm{E}\left[|X|^n\right] = \mathrm{E}\left[X^n\right]$ gives the nth statistical moment of the r.v. X, so that (B.1) provides a series of bounds that involve the higher moments of the r.v..

Another simple bound is provided by the following inequality.

Theorem B.3 (Chebyshev's inequality) *Let X be a random variable with finite expected value m_X and finite non-zero variance σ_X^2. Then for any real number $k > 0$, it holds*

$$\mathrm{P}\left[|X - \mathrm{m}_x| \geq a\right] \leq \frac{\sigma_X^2}{a^2} \tag{B.2}$$

Proof: Let $Y = (X - \mathrm{m}_X)^2$. We can hence write

$$\mathrm{P}\left[|X - \mathrm{m}_x| \geq a\right] = \mathrm{P}\left[(X - \mathrm{m}_x)^2 \geq a^2\right] \leq \frac{\mathrm{E}\left[(X - \mathrm{m}_x)^2\right]}{a^2} = \frac{\sigma_X^2}{a^2}$$

where the inequality follows from Theorem B.1 (Markov inequality). ■

Even though Markov's and Chebyshev's Inequality may be loose for a specific r.v., it is not possible to derive tighter bounds with *general* validity. Indeed, it is possible to find r.v.s for which the Chebyshev's bound is, indeed, tight (left for exercise to the reader).[1]

While it is not possible to improve Markov's Inequality and Chebyshev's Inequality for r.v.s with general distributions, it is instead possible to find stricter bounds for a more restricted, yet interesting, class of r.v.s. In particular, we consider the case of a r.v. that is the sum of a number of other rr.v.s, i.e.,

$$X = \sum_{h=1}^{N} X_h \ ;$$

where $\{X_h\}$ are independent r.v.s with finite mean $\{\mathrm{m}_{X_h}\}$ and variance $\{\sigma^2_{X_h}\}$. We hence have $\mathrm{m}_X = \sum_{h=1}^{N} \mathrm{m}_{X_h}$ and $\sigma^2_X = \sum_{h=1}^{N} \sigma^2_{X_h}$. Then, by applying the Chebyshev's inequality, we get the following result:

$$\mathrm{P}\left[|X - \mathrm{m}_x| \geq a\right] \leq \frac{\sum_{h=1}^{N} \sigma^2_{X_h}}{a^2} \ . \tag{B.3}$$

In particular, if the $\{X_h\}$ are i.i.d. with mean μ and variance σ^2, then (B.3) yields

$$\mathrm{P}\left[\left|\frac{\sum_{h=1}^{N} X_h}{N} - \mu\right| \geq \varepsilon\right] \leq \frac{\sigma^2}{N\varepsilon^2} \ . \tag{B.4}$$

Tighter bounds can be obtained by exploiting the assumption of mutual independence among the r.v.s, or focusing on particular classes of r.v.s. These approaches have been followed by Herman Chernoff, which has derived a number of useful upper and lower bounds (all generally known as *Chernoff bounds*). In the following, we present some of such bounds.

Theorem B.4 (Chernoff bound for iid rvs) *Given a sequence of i.i.d. r.v.s $\{X_h\}$ with mean μ and variance σ^2, for any constant a it holds*

$$\mathrm{P}\left[\frac{1}{N}\sum_{h=1}^{N} X_h \geq a\right] \leq e^{-N \sup_{\theta \geq 0}\{\theta a - \log M(\theta)\}} \tag{B.5}$$

where $M(\theta) = \mathrm{E}\left[e^{\theta X_h}\right]$ is the moment generating function of the r.v.s $\{X_h\}$.

Proof: For any $\theta \geq 0$ we have

$$\mathrm{P}\left[\frac{1}{N}\sum_{h=1}^{N} X_h \geq a\right] \leq \mathrm{P}\left[e^{\theta \sum_{h=1}^{N} X_h} \geq e^{N\theta a}\right] \leq \frac{\mathrm{E}\left[e^{\theta \sum_{h=1}^{N} X_h}\right]}{e^{N\theta a}} = M(\theta)^N e^{-N\theta a} \ , \tag{B.6}$$

where the first inequality is an equality for any non-zero θ, the second follows from the Markov inequality, and the last step is derived by exploiting the independence of the r.v.s

[1] Suggestion: consider a discrete r.v. and show that (B.1) holds with equality at in one specific point of the r.v. measureable space.

and the definition of $M(\theta)$. Rewriting (B.6) we get

$$\mathrm{P}\left[\frac{1}{N}\sum_{h=1}^{N} X_h \geq a\right] \leq \exp\left(-N(\theta a - \log M(\theta))\right) . \tag{B.7}$$

Since (B.9) holds for any $\theta > 0$, then the stricter condition is given by the θ that minimizes the argument of the exponential, which concludes the proof. ∎

Theorem B.5 (Chernoff bound for Bernoulli rvs) *If $X_h \in \{0, 1\}$ are binomial r.v.s, with $\mathrm{P}[X_h = 1] = p$ and $\mathrm{P}[X_h = 0] = 1 - p$, then*

$$\mathrm{P}\left[\frac{1}{N}\sum_{h=1}^{N} X_h \geq a\right] \leq \exp\left(-N D(a||p)\right) . \tag{B.8}$$

where

$$D(a||p) \stackrel{\text{def}}{=} \theta^* a - \log(pe^{\theta^*} + 1 - p) = a \log \frac{a}{p} + (1 - a) \log \frac{1 - a}{1 - p}$$

is the Kullback-Leibler distance[2] between Bernoulli r.v.s with parameters a and p.

Proof: The moment generating function of a Bernoulli r.v. is

$$M(\theta) = \mathrm{E}\left[e^{\theta X_h}\right] = pe^{\theta} + 1 - p\,.$$

The Chernoff bound (B.5), then, gets the form

$$\mathrm{P}\left[\sum_{h=1}^{N} X_h \geq aN\right] \leq e^{-N \sup_{\theta \geq 0}\left\{\theta a - \log(pe^{\theta} + 1 - p)\right\}} , \tag{B.9}$$

which is meaningful only if $0 \leq a < 1$. The value θ^* of θ that maximizes $L(\theta) \stackrel{\text{def}}{=} \theta a - \log(pe^{\theta} + 1 - p)$ is easily obtained by setting to zero the derivative of $L(\theta)$ with respect to θ, and clipping to zero the negative results, which gives

$$\theta^* = \left\{\log\left(\frac{a(1 - p)}{p(1 - a)}\right)\right\}^{+} \tag{B.10}$$

where $\{x\}^{+} = \max(0, x)$. Therefore, if $a \leq p$ we get $\theta^* = 0$ and the bound becomes trivial. Instead, for $a > p$, we have $\theta^* > 0$ and we obtain

$$D(a||p) \stackrel{\text{def}}{=} L(\theta^*) = \theta^* a - \log(pe^{\theta^*} + 1 - p) = a \log \frac{a}{p} + (1 - a) \log \frac{1 - a}{1 - p} ,$$

which concludes the proof. ∎

We can hence see that the tail of the probability decreases at least exponentially fast as N grows to infinity.

[2]We recall that, given two discrete probability functions $P(x)$ and $Q(x)$, their Kullback-Leibler divergence of Q with respect to P is defined as the expectation on P of the difference of the logarithm of the two distributions, i.e., $D(P||Q) = \sum_x P(x) \log(P(x)/Q(x))$.